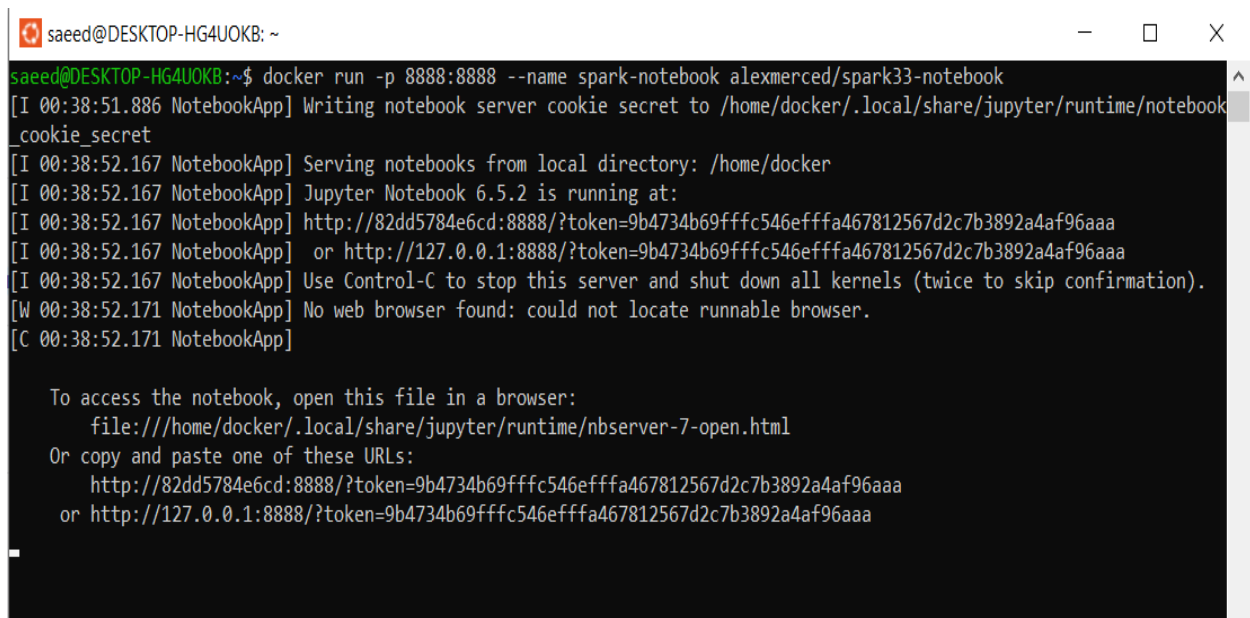


Iceberg Use Case: Banking Transactions Analytics

This project demonstrates how to use Apache Iceberg with PySpark to build a simple data lakehouse for a banking analytics use case.

The goal is to create Iceberg tables for customers, accounts, branches, and transactions, then compute some key KPIs.

Step 1: Run the Spark Notebook Docker Container



```
saeed@DESKTOP-HG4UOKB: ~  
saeed@DESKTOP-HG4UOKB:~$ docker run -p 8888:8888 --name spark-notebook alexmerced/spark33-notebook  
[I 00:38:51.886 NotebookApp] Writing notebook server cookie secret to /home/docker/.local/share/jupyter/runtime/notebook_cookie_secret  
[I 00:38:52.167 NotebookApp] Serving notebooks from local directory: /home/docker  
[I 00:38:52.167 NotebookApp] Jupyter Notebook 6.5.2 is running at:  
[I 00:38:52.167 NotebookApp] http://82dd5784e6cd:8888/?token=9b4734b69fffc546efffa467812567d2c7b3892a4af96aaa  
[I 00:38:52.167 NotebookApp] or http://127.0.0.1:8888/?token=9b4734b69fffc546efffa467812567d2c7b3892a4af96aaa  
[I 00:38:52.167 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).  
[W 00:38:52.171 NotebookApp] No web browser found: could not locate runnable browser.  
[C 00:38:52.171 NotebookApp]  
  
To access the notebook, open this file in a browser:  
file:///home/docker/.local/share/jupyter/runtime/nbserver-7-open.html  
Or copy and paste one of these URLs:  
http://82dd5784e6cd:8888/?token=9b4734b69fffc546efffa467812567d2c7b3892a4af96aaa  
or http://127.0.0.1:8888/?token=9b4734b69fffc546efffa467812567d2c7b3892a4af96aaa
```

This command launches a Jupyter Notebook environment with Apache Spark preinstalled. Copy the URL that appears in the terminal (it includes a token) and open it in your browser

Step 2: Create the Spark Session with Iceberg Configuration

```
In [1]: from pyspark.sql import SparkSession

In [2]: # Iceberg extensions
spark = SparkSession.builder \
    .appName("iceberg-bank-demo") \
    .config("spark.jars.packages", "org.apache.iceberg:iceberg-spark-runtime-3.3_2.12:1.5.2") \
    .config("spark.sql.extensions", "org.apache.iceberg.spark.extensions.IcebergSparkSessionExtensions") \
    .config("spark.sql.catalog.iceberg", "org.apache.iceberg.spark.SparkCatalog") \
    .config("spark.sql.catalog.iceberg.type", "hadoop") \
    .config("spark.sql.catalog.iceberg.warehouse", "iceberg_warehouse") \
    .getOrCreate()

print("Spark with Iceberg is ready")

:: loading settings :: url = jar:file:/home/docker/.local/lib/python3.10/site-packages/pyspark/jars/ivy-2.5.0.jar!/org/apache/ivy/core/settings/ivysettings.xml
```

This initializes Spark with Iceberg support and sets the warehouse directory for storing Iceberg metadata, snapshots, and manifest files.

Step 3: Create the Namespace

```
In [3]: spark.sql("CREATE NAMESPACE IF NOT EXISTS iceberg.bank")
```

```
Out[3]: DataFrame[]
```

```
In [4]: spark.sql("SHOW NAMESPACES").show(truncate=False)
```

```
+-----+
|namespace|
+-----+
|default  |
+-----+
```

This creates a logical namespace (like a database) inside Iceberg called bank.

Step 4: Create the Required Tables

```
In [5]: # Create Customers table
spark.sql("""
CREATE TABLE IF NOT EXISTS iceberg.bank.customers (
    customer_id STRING,
    name STRING,
    age INT,
    gender STRING
)
USING iceberg
""")

# Create Accounts table
spark.sql("""
CREATE TABLE IF NOT EXISTS iceberg.bank.accounts (
    account_id STRING,
    customer_id STRING,
    account_type STRING,
    balance DECIMAL(12,2)
)
USING iceberg
""")

# Create Branches table
spark.sql("""
CREATE TABLE IF NOT EXISTS iceberg.bank.branches (
    branch_id STRING,
    branch_name STRING,
    city STRING
)
USING iceberg
""")

# Create Transactions table
spark.sql("""
CREATE TABLE IF NOT EXISTS iceberg.bank.transactions (
    transaction_id STRING,
    account_id STRING,
    branch_id STRING,
    transaction_date DATE,
    transaction_type STRING,
    amount DECIMAL(12,2)
)
USING iceberg
""")

print("All tables created successfully")
```

All tables created successfully

```
In [6]: spark.sql("SHOW TABLES IN iceberg.bank").show(truncate=False)
```

```
+-----+-----+-----+
|namespace|tableName|isTemporary|
+-----+-----+-----+
|bank      |transactions|false      |
|bank      |branches    |false      |
|bank      |customers   |false      |
|bank      |accounts    |false      |
+-----+-----+-----+
```

Step 5: Insert Sample Data

```
In [7]: spark.sql("""
INSERT INTO iceberg.bank.customers VALUES
('C001', 'Ali Hassan', 35, 'Male'),
('C002', 'Sara Ahmed', 28, 'Female'),
('C003', 'Omar Youssef', 42, 'Male'),
('C004', 'Mona Khaled', 31, 'Female'),
('C005', 'Tarek Nabil', 50, 'Male'),
('C006', 'Layla Mostafa', 24, 'Female')
""")
```

```
Out[7]: DataFrame[]
```

```
In [8]: spark.sql("""
INSERT INTO iceberg.bank.accounts VALUES
('A101', 'C001', 'Checking', 15000.50),
('A102', 'C002', 'Savings', 8000.00),
('A103', 'C003', 'Checking', 22000.75),
('A104', 'C004', 'Savings', 12000.00),
('A105', 'C005', 'Checking', 50000.00),
('A106', 'C006', 'Savings', 3000.00)
""")
```

```
Out[8]: DataFrame[]
```

```
In [9]: spark.sql("""
INSERT INTO iceberg.bank.branches VALUES
('B001', 'Cairo Main', 'Cairo'),
('B002', 'Alex Center', 'Alexandria'),
('B003', 'Giza West', 'Giza'),
('B004', 'Mansoura Central', 'Mansoura'),
('B005', 'Aswan Branch', 'Aswan')
""")
```

```
Out[9]: DataFrame[]
```

```
In [10]: spark.sql("""
INSERT INTO iceberg.bank.transactions VALUES
('T001', 'A101', 'B001', DATE '2025-11-01', 'Deposit', 5000.00),
('T002', 'A102', 'B002', DATE '2025-11-01', 'Withdrawal', 1000.00),
('T003', 'A103', 'B001', DATE '2025-11-02', 'Deposit', 3000.00),
('T004', 'A104', 'B003', DATE '2025-11-03', 'Deposit', 7000.00),
('T005', 'A105', 'B004', DATE '2025-11-03', 'Withdrawal', 2000.00),
('T006', 'A106', 'B002', DATE '2025-11-04', 'Deposit', 1500.00),
('T007', 'A101', 'B001', DATE '2025-11-04', 'Withdrawal', 500.00)
""")
```

```
Out[10]: DataFrame[]
```

Step 6: Calculate KPIs

Total transaction volume per branch

KPIs

```
In [21]: #1. Total transaction volume per branch
spark.sql("""
select b.branch_name, sum(t.amount) as total_transaction_volume
from iceberg.bank.transactions t
join iceberg.bank.branches b
on t.branch_id=b.branch_id
group by b.branch_name
order by total_transaction_volume desc""").show()
```

branch_name	total_transaction_volume
Cairo Main	8500.00
Giza West	7000.00
Alex Center	2500.00
Mansoura Central	2000.00

Top 5 customers by total deposit amount

```
In [33]: #2. Top 5 customers by total deposit amount
spark.sql("""
select c.name , sum(t.amount) as total_deposit_amount
from iceberg.bank.transactions t
join iceberg.bank.accounts a
on t.account_id = a.account_id
join iceberg.bank.customers c
on a.customer_id = c.customer_id
where t.transaction_type = 'Deposit'
group by c.name
order by total_deposit_amount desc
""").show(5)
```

name	total_deposit_amount
Mona Khaled	7000.00
Ali Hassan	5000.00
Omar Youssef	3000.00
Layla Mostafa	1500.00

Average transaction amount per account type

```
In [27]: #3. Average transaction amount per account type
spark.sql("""
select a.account_type, round(avg(t.amount), 2) as avg_transaction_amount
from iceberg.bank.transactions t
join iceberg.bank.accounts a
on t.account_id = a.account_id
group by a.account_type
""").show()
```

account_type	avg_transaction_amount
Checking	2625.00
Savings	3166.67

Number of transactions per day

```
In [26]: #4. Number of transactions per day
spark.sql("""
select transaction_date, count(*) AS num_transactions
from iceberg.bank.transactions
group by transaction_date
order by transaction_date
""").show()
```

transaction_date	num_transactions
2025-11-01	2
2025-11-02	1
2025-11-03	2
2025-11-04	2

Step 7 : Use Snapshots to See Historical Transactions

You can explore Iceberg's versioning and time travel:

Show current snapshots

```
In [34]: spark.sql("SELECT * FROM iceberg.bank.transactions.snapshots").show(truncate=False)
```

```
+-----+-----+-----+-----+-----+
|committed_at|snapshot_id|parent_id|operation|manifest_list|
|summary|
|
+-----+-----+-----+-----+-----+
|2025-11-06 01:50:52.376|8947272129877968963|null|append|iceberg_warehouse/bank/transactions/metadata/snap-8947272129877968963-1-f2240cb8-29a3-49ce-8d95-90b5859160dd.avro|{spark.app.id -> local-1762392942255, added-data-files -> 7, added-records -> 7, added-files-size -> 12284, changed-partition-count -> 1, total-records -> 7, total-files-size -> 12284, total-data-files -> 7, total-delete-files -> 0, total-position-deletes -> 0, total-equality-deletes -> 0}|
+-----+-----+-----+-----+-----+
```

Insert new transaction (creates a new snapshot)

Check new snapshot

```
In [35]: spark.sql("""
INSERT INTO iceberg.bank.transactions VALUES
('T006', 'A003', 'B001', DATE '2025-11-06', 'Deposit', 2000.00)
""")
```

```
Out[35]: DataFrame[]
```

```
In [36]: spark.sql("SELECT * FROM iceberg.bank.transactions.snapshots").show(truncate=False)
```

```
+-----+-----+-----+-----+-----+
|committed_at|snapshot_id|parent_id|operation|manifest_list|
|summary|
|
+-----+-----+-----+-----+-----+
|2025-11-06 01:50:52.376|8947272129877968963|null|append|iceberg_warehouse/bank/transactions/metadata/snap-8947272129877968963-1-f2240cb8-29a3-49ce-8d95-90b5859160dd.avro|{spark.app.id -> local-1762392942255, added-data-files -> 7, added-records -> 7, added-files-size -> 12284, changed-partition-count -> 1, total-records -> 7, total-files-size -> 12284, total-data-files -> 7, total-delete-files -> 0, total-position-deletes -> 0, total-equality-deletes -> 0}|
|2025-11-06 02:26:46.174|2232567780702144297|8947272129877968963|append|iceberg_warehouse/bank/transactions/metadata/snap-2232567780702144297-1-66f26a83-be55-4ca5-b1b7-8d4315b1ba4f.avro|{spark.app.id -> local-1762392942255, added-data-files -> 1, added-records -> 1, added-files-size -> 1746, changed-partition-count -> 1, total-records -> 8, total-files-size -> 14030, total-data-files -> 8, total-delete-files -> 0, total-position-deletes -> 0, total-equality-deletes -> 0}|
+-----+-----+-----+-----+-----+
```

Files
Running
Clusters

Select items to perform actions on them.

0

iceberg_warehouse / bank / transactions / metadata

Upload

New

0

iceberg_warehouse / bank / transactions / metadata

Name

Last Modified

File size

..

seconds ago

66f26a83-be55-4ca5-b1b7-8d4315b1ba4f-m0.avro

31 minutes ago

7 kB

f2240cb8-29a3-49ce-8d95-90b5859160dd-m0.avro

an hour ago

7.18 kB

snap-2232567780702144297-1-66f26a83-be55-4ca5-b1b7-8d4315b1ba4f.avro

31 minutes ago

4.3 kB

snap-8947272129877968963-1-f2240cb8-29a3-49ce-8d95-90b5859160dd.avro

an hour ago

4.23 kB

v1.metadata.json

an hour ago

1.47 kB

v2.metadata.json

an hour ago

2.5 kB

v3.metadata.json

31 minutes ago

3.47 kB

version-hint.text

31 minutes ago

1 B

```

54  },
55  "properties" : {
56    "owner" : "docker",
57    "write.parquet.compression.codec" : "zstd"
58  },
59  "current-snapshot-id" : 2232567780702144297,
60  "refs" : {
61    "main" : {
62      "snapshot-id" : 2232567780702144297,
63      "type" : "branch"
64    }
65  },
66  "snapshots" : [ {
67    "sequence-number" : 1,
68    "snapshot-id" : 8947272129877968963,
69    "timestamp-ms" : 1762393852376,
70    "summary" : {
71      "operation" : "append",
72      "spark.app.id" : "local-1762392942255",
73      "added-data-files" : "7",
74      "added-records" : "7",
75      "added-files-size" : "12284",
76      "changed-partition-count" : "1",
77      "total-records" : "7",
78      "total-files-size" : "12284",
79      "total-data-files" : "7",
80      "total-delete-files" : "0",
81      "total-position-deletes" : "0",
82      "total-equality-deletes" : "0"
83    },
84    "manifest-list" : "iceberg_warehouse/bank/transactions/metadata/snap-8947272129877968963-1-f2240cb8-29a3-49ce-8d95-90b5859160dd.avro",
85    "schema-id" : 0
86  }, {
87    "sequence-number" : 2,
88    "snapshot-id" : 2232567780702144297,
89    "parent-snapshot-id" : 8947272129877968963,
90    "timestamp-ms" : 1762396006174.

```

View data from an old snapshot

In [38]: spark.read.option("snapshot-id", "8947272129877968963").table("iceberg.bank.transactions").show()

transaction_id	account_id	branch_id	transaction_date	transaction_type	amount
T001	A101	B001	2025-11-01	Deposit	5000.00
T002	A102	B002	2025-11-01	Withdrawal	1000.00
T003	A103	B001	2025-11-02	Deposit	3000.00
T004	A104	B003	2025-11-03	Deposit	7000.00
T005	A105	B004	2025-11-03	Withdrawal	2000.00
T006	A106	B002	2025-11-04	Deposit	1500.00
T007	A101	B001	2025-11-04	Withdrawal	500.00

This completes the **Banking Transactions Analytics with Iceberg** use case, showing how to:

- Configure Spark with Iceberg
- Create and populate tables
- Run analytical SQL queries
- Use snapshots for historical tracking and data versioning